

## LAB ASSIGNMENT-02

Q1. What are the different protocols you observe at the following layers of the protocol stack?

A. Application layer

- HTTP
- HTTPS (TLS)
- DNS

B. Transport Layer

- TCP
- UDP

C. Network Layer

- IPv4

2. What is the total amount of data being received for the following two cases?

A. When you access <http://neverssl.com/>

When accessing:- <http://neverssl.com>

- Total data received:  $\approx 8,821$  bytes

B. When you access <https://www.cornell.edu>

When accessing:- <https://www.cornell.edu>

- Total data received:  $\approx 9,632$  bytes

3. Answer the following when you access the site <http://neverssl.com/>.

A. How many HTTP GET requests do you observe? List down the GET requests

→ HTTP GET requests observed

A total of 4 HTTP GET requests:

1. GET / (Host: neverssl.com)
2. GET / (Host: wholeyoungoldmelody.neverssl.com)
3. GET /online/
4. GET /favicon.ico

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3. B. What version of HTTP the server is running?

HTTP version:- HTTP/1.1

C. For each of the HTTP GET requests, find out:

i) The total number of TCP segments being received  
→ 2 TCP segments

ii) The total amount of data being received in the corresponding HTTP response message.

→ 3027 bytes of data being received.

D. Identify TCP three way handshake used to establish the connection with the `http://neverssl.com`. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each.

Soln → TCP Three-Way Handshake

For one successful HTTP connection:

| <u>Packet</u> | <u>Sequence Number</u> |
|---------------|------------------------|
|---------------|------------------------|

|     |            |
|-----|------------|
| SYN | 4164423771 |
|-----|------------|

|         |           |
|---------|-----------|
| SYN-ACK | 304036046 |
|---------|-----------|

|     |                                                     |
|-----|-----------------------------------------------------|
| ACK | Client acknowledges 304036047<br>(Seq = 4164423772) |
|-----|-----------------------------------------------------|